

Raw scores to standard scores Ages 8 through 9					
Learning Raw	Spoken Words	Written Words	Objects	Designs	Pseudo-words
1	58	67	61	64	81
2	59	68	61	65	82
3	60	69	62	66	82
4	61	69	63	67	83
5	61	70	63	68	84
6	62	71	64	69	84
7	63	72	65	70	85
8	64	72	66	71	86
9	64	73	66	72	87
10	65	74	67	73	87
11	66	74	68	74	88
12	67	75	68	75	89
13	68	76	69	76	90
14	68	76	70	77	90
15	69	77	71	78	91
16	70	78	71	79	92
17	71	78	72	80	92
18	72	79	73	81	93
19	72	80	74	82	94
20	73	80	74	83	95
21	74	81	75	84	95
22	75	82	76	85	96
23	75	82	76	86	97
24	76	83	77	87	97
25	77	84	78	88	98
26	78	84	79	89	99
27	79	85	79	90	100
28	79	86	80	91	100
29	80	86	81	92	101
30	81	87	81	93	102

Raw scores to standard scores Ages 8 through 9					
Learning Raw	Spoken Words	Written Words	Objects	Designs	Pseudo-words
31	82	88	82	94	102
32	83	88	83	95	103
33	83	89	84	96	104
34	84	90	84	97	105
35	85	91	85	98	105
36	86	91	86	99	106
37	86	92	86	101	107
38	87	93	87	102	108
39	88	93	88	103	108
40	89	94	89	104	109
41	90	95	89	105	110
42	90	95	90	106	110
43	91	96	91	107	111
44	92	97	91	108	112
45	93	97	92	109	113
46	94	98	93	110	113
47	94	99	94	111	114
48	95	99	94	112	115
49	96	100	95	113	115
50	97	101	96	114	116
51	98	101	97	115	117
52	98	102	97	116	118
53	99	103	98	117	118
54	100	103	99	118	119
55	101	104	99	119	120
56	101	105	100	120	120
57	102	105	101	121	121
58	103	106	102	122	122
59	104	107	102	123	123
60	105	108	103	124	123

Raw scores to standard scores Ages 8 through 9					
Learning Raw	Spoken Words	Written Words	Objects	Designs	Pseudo-words
61	105	108	104	125	124
62	106	109	104	126	125
63	107	110	105	127	125
64	108	110	106	128	126
65	109	111	107	129	127
66	109	112	107	130	128
67	110	112	108	131	128
68	111	113	109	132	129
69	112	114	109	133	130
70	112	114	110	134	131
71	113	115	111	135	131
72	114	116	112	136	132
73	115	116	112	137	133
74	116	117	113	139	133
75	116	118	114	140	134
76	117	118	115	141	135
77	118	119	115	142	136
78	119	120	116	143	136
79	120	120	117	144	137
80	120	121	117	145	138
82	121	122	118	146	138
83	122	122	119	147	139
84	123	123	120	148	140
85	123	124	120	149	141
86	124	124	121	150	141
87	125	125	122	151	142
88	126	126	122	152	143
89	127	127	123	153	143
90	127	127	124	154	144
91	128	128	125	155	145

Raw scores to standard scores Ages 8 through 9					
Learning Raw	Spoken Words	Written Words	Objects	Designs	Pseudo-words
92	130	129	126	157	146
93	131	130	127	158	147
94	131	131	127	159	148
95	132	131	128	160	149
96	133	132	129	161	149
97	134	133	130	162	150
98	134	133	130	163	151
99	135	134	131	164	151
100	136	135	132	165	152

Learning Gain

Ages 8 through 9

Raw scores to standard scores Ages 8 through 9					
Learning gain Raw	Spoken Words	Written Words	Objects	Designs	Pseudo-words
-30	36	53	60	51	59
-29	37	54	61	52	60
-28	38	55	61	53	61
-27	39	55	62	54	61
-26	39	56	63	55	62
-25	40	57	63	56	63
-24	41	57	64	57	64
-23	42	58	65	58	64
-22	43	59	65	59	65
-21	44	59	66	60	66
-20	45	60	67	61	67
-19	46	61	68	61	67
-18	47	61	68	62	68
-17	48	62	69	63	69
-16	49	63	70	64	69
-15	50	63	70	65	70
-14	51	64	71	66	71
-13	52	65	72	67	72
-12	53	65	73	68	72
-11	54	66	73	69	73
-10	54	67	74	70	74
-9	55	67	75	71	75
-8	56	68	75	72	75
-7	57	69	76	73	76
-6	58	70	77	74	77
-5	59	70	77	75	78
-4	60	71	78	76	78
-3	61	72	79	77	79
-2	62	72	80	78	80
-1	63	73	80	79	81

Raw scores to standard scores Ages 8 through 9					
Learning gain Raw	Spoken Words	Written Words	Objects	Designs	Pseudo-words
0	64	74	81	80	81
1	65	74	82	81	82
2	66	75	82	82	83
3	67	76	83	83	84
4	68	76	84	84	84
5	69	77	85	85	85
6	70	78	85	86	86
7	70	78	86	86	86
8	71	79	87	87	87
9	72	80	87	88	88
10	73	80	88	89	89
11	74	81	89	90	89
12	75	82	89	91	90
13	76	82	90	92	91
14	77	83	91	93	92
15	78	84	92	94	92
16	79	84	92	95	93
17	80	85	93	96	94
18	81	86	94	97	95
19	82	86	94	98	95
20	83	87	95	99	96
21	84	88	96	100	97
22	85	88	96	101	98
23	85	89	97	102	98
24	86	90	98	103	99
25	87	90	99	104	100
26	88	91	99	105	100
27	89	92	100	106	101
28	90	92	101	107	102
29	91	93	101	108	103

Learning Gain
Ages 8 through 9

Raw scores to standard scores Ages 8 through 9					
Learning gain Raw	Spoken Words	Written Words	Objects	Designs	Pseudo-words
30	92	94	102	109	103
31	93	94	103	110	104
32	94	95	104	111	105
33	95	96	104	111	106
34	96	96	105	112	106
35	97	97	106	113	107
36	98	98	106	114	108
37	99	98	107	115	109
38	100	99	108	116	109
39	100	100	108	117	110
40	101	100	109	118	111
41	102	101	110	119	112
42	103	102	111	120	112
43	104	102	111	121	113
44	105	103	112	122	114
45	106	104	113	123	115
46	107	104	113	124	115
47	108	105	114	125	116
48	109	106	115	126	117
49	110	106	115	127	117
50	111	107	116	128	118
51	112	108	117	129	119
52	113	108	118	130	120
53	114	109	118	131	120
54	115	110	119	132	121
55	115	110	120	133	122
56	116	111	120	134	123
57	117	112	121	135	123
58	118	112	122	136	124
59	119	113	123	136	125

Raw scores to standard scores Ages 8 through 9					
Learning gain Raw	Spoken Words	Written Words	Objects	Designs	Pseudo-words
60	120	114	123	137	126
61	121	114	124	138	126
62	122	115	125	139	127
63	123	116	125	140	128
64	124	116	126	141	129
65	125	117	127	142	129
66	126	118	127	143	130
67	127	118	128	144	131
68	128	119	129	145	132
69	129	120	130	146	132
70	130	121	130	147	133
71	130	121	131	148	134
72	131	122	132	149	134
73	132	123	132	150	135
74	133	123	133	151	136
75	134	124	134	152	137

Delayed recall

Ages 8 through 9

Raw scores to standard scores Ages 8 through 9					
Delayed recall Raw	Spoken Words	Written Words	Objects	Designs	Pseudo-words
0	59	73	61	66	79
1	60	73	62	66	80
2	61	74	62	67	80
3	61	74	63	68	81
4	62	75	64	69	82
5	62	75	64	70	82
6	63	76	65	70	83
7	64	76	65	71	84
8	64	77	66	72	84
9	65	77	67	73	85
10	66	78	67	74	86
11	66	78	68	74	86
12	67	78	69	75	87
13	67	79	69	76	88
14	68	79	70	77	88
15	69	80	70	78	89
16	69	80	71	78	89
17	70	81	72	79	90
18	71	81	72	80	91
19	71	82	73	81	91
20	72	82	73	82	92
21	72	83	74	82	93
22	73	83	75	83	93
23	74	84	75	84	94
24	74	84	76	85	95
25	75	85	76	86	95
26	76	85	77	86	96
27	76	85	78	87	97
28	77	86	78	88	97
29	77	86	79	89	98

Raw scores to standard scores Ages 8 through 9					
Delayed recall Raw	Spoken Words	Written Words	Objects	Designs	Pseudo-words
30	78	87	80	90	99
31	79	87	80	90	99
32	79	88	81	91	100
33	80	88	81	92	101
34	81	89	82	93	101
35	81	89	83	94	102
36	82	90	83	94	103
37	82	90	84	95	103
38	83	91	84	96	104
39	84	91	85	97	105
40	84	91	86	97	105
41	85	92	86	98	106
42	86	92	87	99	107
43	86	93	87	100	107
44	87	93	88	101	108
45	87	94	89	101	109
46	88	94	89	102	109
47	89	95	90	103	110
48	89	95	91	104	111
49	90	96	91	105	111
50	91	96	92	105	112
51	91	97	92	106	112
52	92	97	93	107	113
53	92	97	94	108	114
54	93	98	94	109	114
55	94	98	95	109	115
56	94	99	95	110	116
57	95	99	96	111	116
58	96	100	97	112	117
59	96	100	97	113	118

Delayed recall

Ages 8 through 9

Raw scores to standard scores Ages 8 through 9					
Delayed recall Raw	Spoken Words	Written Words	Objects	Designs	Pseudo-words
60	97	101	98	113	118
61	97	101	98	114	119
62	98	102	99	115	120
63	99	102	100	116	120
64	99	103	100	117	121
65	100	103	101	117	122
66	101	104	102	118	122
67	101	104	102	119	123
68	102	104	103	120	124
69	103	105	103	121	124
70	103	105	104	121	125
71	104	106	105	122	126
72	104	106	105	123	126
73	105	107	106	124	127
74	106	107	106	125	128
75	106	108	107	125	128
76	107	108	108	126	129
77	108	109	108	127	130
78	108	109	109	128	130
79	109	110	109	128	131
80	109	110	110	129	132
81	110	110	111	130	132
82	111	111	111	131	133
83	111	111	112	132	133
84	112	112	113	132	134
85	113	112	113	133	135
86	113	113	114	134	135
87	114	113	114	135	136
88	114	114	115	136	137
89	115	114	116	136	137

Raw scores to standard scores Ages 8 through 9					
Delayed recall Raw	Spoken Words	Written Words	Objects	Designs	Pseudo-words
90	116	115	116	137	138
91	116	115	117	138	139
92	117	116	117	139	139
93	118	116	118	140	140
94	118	116	119	140	141
95	119	117	119	141	141
96	119	117	120	142	142
97	120	118	120	143	143
98	121	118	121	144	143
99	121	119	122	144	144
100	122	119	122	145	145

Forgetting
Ages 8 through 9

Raw scores to standard scores Ages 8 through 9					
Forgetting Raw	Spoken Words	Written Words	Objects	Designs	Pseudo-words
-50	12	43	39	38	45
-49	14	45	40	40	46
-48	16	46	41	41	47
-47	18	47	43	42	49
-46	20	49	44	44	50
-45	22	50	45	45	51
-44	25	52	47	46	52
-43	27	53	48	48	54
-42	29	54	49	49	55
-41	31	56	51	50	56
-40	33	57	52	52	57
-39	35	59	53	53	58
-38	37	60	54	54	60
-37	39	62	56	56	61
-36	41	63	57	57	62
-35	43	64	58	58	63
-34	45	66	60	60	65
-33	47	67	61	61	66
-32	49	69	62	62	67
-31	51	70	64	64	68
-30	53	72	65	65	70
-29	55	73	66	66	71
-28	57	74	68	67	72
-27	59	76	69	69	73
-26	61	77	70	70	74
-25	63	79	72	71	76
-24	65	80	73	73	77
-23	67	82	74	74	78
-22	69	83	76	75	79
-21	71	84	77	77	81

Raw scores to standard scores Ages 8 through 9					
Forgetting Raw	Spoken Words	Written Words	Objects	Designs	Pseudo-words
-20	73	86	78	78	82
-19	75	87	80	79	83
-18	77	89	81	81	84
-17	79	90	82	82	86
-16	81	91	84	83	87
-15	83	93	85	85	88
-14	85	94	86	86	89
-13	87	96	87	87	90
-12	89	97	89	89	92
-11	91	99	90	90	93
-10	93	100	91	91	94
-9	95	101	93	93	95
-8	97	103	94	94	97
-7	99	104	95	95	98
-6	101	106	97	96	99
-5	103	107	98	98	100
-4	105	109	99	99	102
-3	107	110	101	100	103
-2	109	111	102	102	104
-1	111	113	103	103	105
0	113	114	105	104	106
1	115	116	106	106	108
2	117	117	107	107	109
3	119	118	109	108	110
4	121	120	110	110	111
5	123	121	111	111	113
6	125	123	113	112	114
7	127	124	114	114	115
8	129	126	115	115	116
9	131	127	116	116	118

Raw scores to standard scores Ages 8 through 9					
Forgetting Raw	Spoken Words	Written Words	Objects	Designs	Pseudo- words
10	133	128	118	118	119
11	135	130	119	119	120
12	137	131	120	120	121
13	139	133	122	122	122
14	141	134	123	123	124
15	143	136	124	124	125
16	145	137	126	125	126
17	147	138	127	127	127
18	149	140	128	128	129
19	151	141	130	129	130

Recognition

Ages 8 through 9

Raw scores to standard scores Ages 8 through 9					
Recognition Raw	Spoken Words	Written Words	Objects	Designs	Pseudo- words
-30	52	47	29	60	66
-29	53	48	30	60	67
-28	53	48	31	61	67
-27	54	49	31	61	68
-26	54	49	32	62	68
-25	55	50	32	62	68
-24	55	50	33	62	69
-23	55	51	34	63	69
-22	56	51	34	63	69
-21	56	52	35	64	70
-20	57	52	36	64	70
-19	57	53	36	65	70
-18	58	53	37	65	71
-17	58	54	38	65	71
-16	58	55	38	66	71
-15	59	55	39	66	72
-14	59	56	39	67	72
-13	60	56	40	67	72
-12	60	57	41	68	73
-11	61	57	41	68	73
-10	61	58	42	68	74
-9	62	58	43	69	74
-8	62	59	43	69	74
-7	62	59	44	70	75
-6	63	60	45	70	75
-5	63	60	45	71	75
-4	64	61	46	71	76
-3	64	61	46	71	76
-2	65	62	47	72	76
-1	65	62	48	72	77

Raw scores to standard scores Ages 8 through 9					
Recognition Raw	Spoken Words	Written Words	Objects	Designs	Pseudo- words
0	66	63	48	73	77
1	66	63	49	73	77
2	66	64	50	74	78
3	67	64	50	74	78
4	67	65	51	75	78
5	68	65	52	75	79
6	68	66	52	75	79
7	69	66	53	76	79
8	69	67	53	76	80
9	70	67	54	77	80
10	70	68	55	77	81
11	70	68	55	78	81
12	71	69	56	78	81
13	71	69	57	78	82
14	72	70	57	79	82
15	72	70	58	79	82
16	73	71	59	80	83
17	73	71	59	80	83
18	74	72	60	81	83
19	74	72	60	81	84
20	74	73	61	81	84
21	75	73	62	82	84
22	75	74	62	82	85
23	76	75	63	83	85
24	76	75	64	83	85
25	77	76	64	84	86
26	77	76	65	84	86
27	77	77	66	84	87
28	78	77	66	85	87
29	78	78	67	85	87

Recognition

Ages 8 through 9

Raw scores to standard scores Ages 8 through 9					
Recognition Raw	Spoken Words	Written Words	Objects	Designs	Pseudo- words
30	79	78	67	86	88
31	79	79	68	86	88
32	80	79	69	87	88
33	80	80	69	87	89
34	81	80	70	87	89
35	81	81	71	88	89
36	81	81	71	88	90
37	82	82	72	89	90
38	82	82	73	89	90
39	83	83	73	90	91
40	83	83	74	90	91
41	84	84	74	90	91
42	84	84	75	91	92
43	85	85	76	91	92
44	85	85	76	92	93
45	85	86	77	92	93
46	86	86	78	93	93
47	86	87	78	93	94
48	87	87	79	93	94
49	87	88	80	94	94
50	88	88	80	94	95
51	88	89	81	95	95
52	89	89	81	95	95
53	89	90	82	96	96
54	89	90	83	96	96
55	90	91	83	96	96
56	90	91	84	97	97
57	91	92	85	97	97
58	91	92	85	98	97
59	92	93	86	98	98

Raw scores to standard scores Ages 8 through 9					
Recognition Raw	Spoken Words	Written Words	Objects	Designs	Pseudo- words
60	92	94	87	99	98
61	93	94	87	99	98
62	93	95	88	100	99
63	93	95	88	100	99
64	94	96	89	100	100
65	94	96	90	101	100
66	95	97	90	101	100
67	95	97	91	102	101
68	96	98	92	102	101
69	96	98	92	103	101
70	97	99	93	103	102
71	97	99	94	103	102
72	97	100	94	104	102
73	98	100	95	104	103
74	98	101	96	105	103
75	99	101	96	105	103
76	99	102	97	106	104
77	100	102	97	106	104
78	100	103	98	106	104
79	100	103	99	107	105
80	101	104	99	107	105
81	101	104	100	108	106
82	102	105	101	108	106
83	102	105	101	109	106
84	103	106	102	109	107
85	103	106	103	109	107
86	104	107	103	110	107
87	104	107	104	110	108
88	104	108	104	111	108
89	105	108	105	111	108

Raw scores to standard scores Ages 8 through 9					
Recognition Raw	Spoken Words	Written Words	Objects	Designs	Pseudo- words
90	105	109	106	112	109
91	106	109	106	112	109
92	106	110	107	112	109
93	107	110	108	113	110
94	107	111	108	113	110
95	108	111	109	114	110
96	108	112	110	114	111
97	108	113	110	115	111
98	109	113	111	115	112
99	109	114	111	115	112
100	110	114	112	116	112

Raw scores to standard scores Ages 10 through 11					
Learning Raw	Spoken Words	Written Words	Objects	Designs	Pseudo-words
1	65	69	67	72	72
2	66	70	67	73	73
3	66	70	68	74	74
4	67	71	69	75	75
5	68	72	69	75	76
6	68	72	70	76	76
7	69	73	70	77	77
8	70	73	71	78	78
9	70	74	72	79	79
10	71	75	72	80	80
11	72	75	73	81	81
12	72	76	74	82	82
13	73	76	74	82	83
14	73	77	75	83	84
15	74	78	76	84	85
16	75	78	76	85	85
17	75	79	77	86	86
18	76	80	78	87	87
19	77	80	78	88	88
20	77	81	79	88	89
21	78	81	79	89	90
22	79	82	80	90	91
23	79	83	81	91	92
24	80	83	81	92	93
25	81	84	82	93	94
26	81	84	83	94	95
27	82	85	83	95	95
28	83	86	84	95	96
29	83	86	85	96	97
30	84	87	85	97	98

Raw scores to standard scores Ages 10 through 11					
Learning Raw	Spoken Words	Written Words	Objects	Designs	Pseudo-words
31	84	87	86	98	99
32	85	88	87	99	100
33	86	89	87	100	101
34	86	89	88	101	102
35	87	90	88	101	103
36	88	91	89	102	104
37	88	91	90	103	104
38	89	92	90	104	105
39	90	92	91	105	106
40	90	93	92	106	107
41	91	94	92	107	108
42	92	94	93	107	109
43	92	95	94	108	110
44	93	95	94	109	111
45	93	96	95	110	112
46	94	97	96	111	113
47	95	97	96	112	114
48	95	98	97	113	114
49	96	98	97	114	115
50	97	99	98	114	116
51	97	100	99	115	117
52	98	100	99	116	118
53	99	101	100	117	119
54	99	102	101	118	120
55	100	102	101	119	121
56	101	103	102	120	122
57	101	103	103	120	123
58	102	104	103	121	123
59	102	105	104	122	124
60	103	105	105	123	125

Raw scores to standard scores Ages 10 through 11					
Learning Raw	Spoken Words	Written Words	Objects	Designs	Pseudo-words
61	104	106	105	124	126
62	104	106	106	125	126
63	105	107	106	126	127
64	106	108	107	127	128
65	106	108	108	127	129
66	107	109	108	128	130
67	108	109	109	129	131
68	108	110	110	130	132
69	109	111	110	131	133
70	110	111	111	132	133
71	110	112	112	133	134
72	111	113	112	133	135
73	111	113	113	134	136
74	112	114	114	135	137
75	113	114	114	136	138
76	113	115	115	137	139
77	114	116	115	138	140
78	115	116	116	139	141
79	115	117	117	140	141
80	116	117	117	140	142
81	117	118	118	141	143
82	117	119	119	142	144
83	118	119	119	143	145
84	119	120	120	144	146
85	119	120	121	145	147
86	120	121	121	146	148
87	120	122	122	146	149
88	121	122	123	147	149
89	122	123	123	148	150
90	122	124	124	149	151

Raw scores to standard scores Ages 10 through 11					
Learning Raw	Spoken Words	Written Words	Objects	Designs	Pseudo-words
91	123	124	124	150	153
92	124	125	125	151	154
93	124	125	126	152	155
94	125	126	126	152	156
95	126	127	127	153	157
96	126	127	128	154	158
97	127	128	128	155	159
98	128	128	129	156	160
99	128	129	130	157	161
100	129	130	130	158	161

Raw scores to standard scores Ages 10 through 11					
Learning gain Raw	Spoken Words	Written Words	Objects	Designs	Pseudo- words
-30	56	56	62	38	62
-29	57	56	62	39	62
-28	57	57	63	40	63
-27	58	58	64	42	64
-26	59	58	64	43	64
-25	59	59	65	44	65
-24	60	60	66	45	65
-23	61	60	66	46	66
-22	62	61	67	48	67
-21	62	62	68	49	67
-20	63	62	69	50	68
-19	64	63	69	51	69
-18	64	64	70	52	69
-17	65	64	71	54	70
-16	66	65	71	55	71
-15	66	65	72	56	71
-14	67	66	73	57	72
-13	68	67	73	58	73
-12	69	67	74	60	73
-11	69	68	75	61	74
-10	70	69	75	62	75
-9	71	69	76	63	75
-8	71	70	77	64	76
-7	72	71	77	66	77
-6	73	71	78	67	77
-5	73	72	79	68	78
-4	74	73	79	69	79
-3	75	73	80	70	79
-2	76	74	81	72	80
-1	76	75	81	73	80

Raw scores to standard scores Ages 10 through 11					
Learning gain Raw	Spoken Words	Written Words	Objects	Designs	Pseudo- words
0	77	75	82	74	81
1	78	76	83	75	82
2	78	77	83	76	82
3	79	77	84	78	83
4	80	78	85	79	84
5	80	79	85	80	84
6	81	79	86	81	85
7	82	80	87	82	86
8	83	81	87	84	86
9	83	81	88	85	87
10	84	82	89	86	88
11	85	83	89	87	88
12	85	83	90	88	89
13	86	84	91	90	90
14	87	85	91	91	90
15	88	85	92	92	91
16	88	86	93	93	92
17	89	87	93	94	92
18	90	87	94	96	93
19	90	88	95	97	94
20	91	89	96	98	94
21	92	89	96	99	95
22	92	90	97	100	96
23	93	91	98	102	96
24	94	91	98	103	97
25	95	92	99	104	97
26	95	93	100	105	98
27	96	93	100	106	99
28	97	94	101	108	99
29	97	95	102	109	100

Raw scores to standard scores Ages 10 through 11					
Learning gain Raw	Spoken Words	Written Words	Objects	Designs	Pseudo- words
30	98	95	102	110	101
31	99	96	103	111	101
32	99	97	104	112	102
33	100	97	104	114	103
34	101	98	105	115	103
35	102	99	106	116	104
36	102	99	106	117	105
37	103	100	107	118	105
38	104	101	108	120	106
39	104	101	108	121	107
40	105	102	109	122	107
41	106	102	110	123	108
42	106	103	110	124	109
43	107	104	111	126	109
44	108	104	112	127	110
45	109	105	112	128	111
46	109	106	113	129	111
47	110	106	114	130	112
48	111	107	114	132	112
49	111	108	115	133	113
50	112	108	116	134	114
51	113	109	116	135	114
52	113	110	117	136	115
53	114	110	118	138	116
54	115	111	118	139	116
55	116	112	119	140	117
56	116	112	120	141	118
57	117	113	120	142	118
58	118	114	121	144	119
59	118	114	122	145	120

Raw scores to standard scores Ages 10 through 11					
Learning gain Raw	Spoken Words	Written Words	Objects	Designs	Pseudo- words
60	119	115	122	146	120
61	120	116	123	147	121
62	121	116	124	148	122
63	121	117	125	150	122
64	122	118	125	151	123
65	123	118	126	152	124
66	123	119	127	153	124
67	124	120	127	154	125
68	125	120	128	156	126
69	125	121	129	157	126
70	126	122	129	158	127
71	127	122	130	159	127
72	128	123	131	160	128
73	128	124	131	162	129
74	129	124	132	163	129
75	130	125	133	164	130
76	130	126	133	165	131
77	131	126	134	166	131
78	132	127	135	168	132
79	132	128	135	169	133
80	133	128	136	170	133
81	134	129	137	171	134
82	135	130	137	172	135
83	135	130	138	174	135

Delayed recall

Ages 10 through 11

Raw scores to standard scores Ages 10 through 11					
Delayed recall	Spoken Words	Written Words	Objects	Designs	Pseudo-words
Raw					
0	63	69	68	72	78
1	64	69	69	72	79
2	64	70	69	73	80
3	65	70	70	74	80
4	66	71	70	74	81
5	66	71	71	75	81
6	67	72	71	76	82
7	67	72	72	76	83
8	68	73	72	77	83
9	68	73	73	78	84
10	69	74	73	78	85
11	69	74	74	79	85
12	70	75	74	80	86
13	71	75	75	80	86
14	71	76	76	81	87
15	72	76	76	82	88
16	72	77	77	82	88
17	73	77	77	83	89
18	73	78	78	84	90
19	74	78	78	84	90
20	74	79	79	85	91
21	75	79	79	86	91
22	76	80	80	86	92
23	76	81	80	87	93
24	77	81	81	88	93
25	77	82	81	88	94
26	78	82	82	89	94
27	78	83	82	90	95
28	79	83	83	90	96
29	79	84	83	91	96

Raw scores to standard scores Ages 10 through 11					
Delayed recall	Spoken Words	Written Words	Objects	Designs	Pseudo-words
Raw					
30	80	84	84	92	97
31	81	85	84	92	98
32	81	85	85	93	98
33	82	86	85	94	99
34	82	86	86	94	99
35	83	87	86	95	100
36	83	87	87	96	101
37	84	88	87	96	101
38	84	88	88	97	102
39	85	89	89	98	102
40	86	89	89	99	103
41	86	90	90	99	104
42	87	90	90	100	104
43	87	91	91	101	105
44	88	91	91	101	106
45	88	92	92	102	106
46	89	92	92	103	107
47	89	93	93	103	107
48	90	93	93	104	108
49	91	94	94	105	109
50	91	94	94	105	109
51	92	95	95	106	110
52	92	95	95	107	110
53	93	96	96	107	111
54	93	96	96	108	112
55	94	97	97	109	112
56	94	97	97	109	113
57	95	98	98	110	114
58	96	98	98	111	114
59	96	99	99	111	115

Raw scores to standard scores Ages 10 through 11					
Delayed recall	Spoken Words	Written Words	Objects	Designs	Pseudo-words
Raw					
60	97	99	99	112	115
61	97	100	100	113	116
62	98	100	100	113	117
63	98	101	101	114	117
64	99	101	102	115	118
65	99	102	102	115	119
66	100	102	103	116	119
67	100	103	103	117	120
68	101	103	104	117	120
69	102	104	104	118	121
70	102	104	105	119	122
71	103	105	105	119	122
72	103	105	106	120	123
73	104	106	106	121	123
74	104	106	107	121	124
75	105	107	107	122	125
76	105	107	108	123	125
77	106	108	108	123	126
78	107	108	109	124	127
79	107	109	109	125	127
80	108	109	110	125	128
81	108	110	110	126	128
82	109	110	111	127	129
83	109	111	111	127	130
84	110	111	112	128	130
85	110	112	112	129	131
86	111	112	113	129	131
87	112	113	113	130	132
88	112	113	114	131	133
89	113	114	114	131	133

Raw scores to standard scores Ages 10 through 11					
Delayed recall	Spoken Words	Written Words	Objects	Designs	Pseudo-words
Raw					
90	113	114	115	132	134
91	114	115	116	133	135
92	114	115	116	133	135
93	115	116	117	134	136
94	115	116	117	135	136
95	116	117	118	135	137
96	117	117	118	136	138
97	117	118	119	137	138
98	118	118	119	137	139
99	118	119	120	138	139
100	119	119	120	139	140

Forgetting
Ages 10 through 11

Raw scores to standard scores Ages 10 through 11					
Forgetting Raw	Spoken Words	Written Words	Objects	Designs	Pseudo-words
-50	17	37	58	28	44
-49	18	38	59	30	45
-48	20	39	60	31	47
-47	22	41	61	33	48
-46	24	42	62	34	49
-45	26	44	63	36	51
-44	27	45	64	37	52
-43	29	47	65	39	54
-42	31	48	66	40	55
-41	33	50	67	41	57
-40	35	51	68	43	58
-39	36	52	69	44	60
-38	38	54	70	46	61
-37	40	55	71	47	63
-36	42	57	72	49	64
-35	43	58	73	50	66
-34	45	60	73	52	67
-33	47	61	74	53	69
-32	49	63	75	55	70
-31	51	64	76	56	72
-30	52	65	77	58	73
-29	54	67	78	59	75
-28	56	68	79	61	76
-27	58	70	80	62	78
-26	60	71	81	63	79
-25	61	73	82	65	81
-24	63	74	83	66	82
-23	65	76	84	68	84
-22	67	77	85	69	85
-21	68	78	86	71	87

Raw scores to standard scores Ages 10 through 11					
Forgetting Raw	Spoken Words	Written Words	Objects	Designs	Pseudo-words
-20	70	80	87	72	88
-19	72	81	88	74	90
-18	74	83	89	75	91
-17	76	84	90	77	93
-16	77	86	91	78	94
-15	79	87	92	80	96
-14	81	89	92	81	97
-13	83	90	93	83	99
-12	85	91	94	84	100
-11	86	93	95	85	102
-10	88	94	96	87	103
-9	90	96	97	88	105
-8	92	97	98	90	106
-7	93	99	99	91	108
-6	95	100	100	93	109
-5	97	102	101	94	111
-4	99	103	102	96	112
-3	101	105	103	97	114
-2	102	106	104	99	115
-1	104	107	105	100	117
0	106	109	106	102	118
1	108	110	107	103	120
2	110	112	108	105	121
3	111	113	109	106	123
4	113	115	110	107	124
5	115	116	111	109	126
6	117	118	112	110	127
7	118	119	112	112	129
8	120	120	113	113	130
9	122	122	114	115	132

Raw scores to standard scores Ages 10 through 11					
Forgetting Raw	Spoken Words	Written Words	Objects	Designs	Pseudo- words
10	124	123	115	116	133
11	126	125	116	118	135
12	127	126	117	119	136
13	129	128	118	121	138
14	131	129	119	122	139
15	133	131	120	124	141
16	135	132	121	125	142
17	136	133	122	127	144
18	138	135	123	128	145
19	140	136	124	129	147
20	142	138	125	131	148
21	143	139	126	132	150
22	145	141	127	134	151
23	147	142	128	135	153
24	149	144	129	137	154
25	151	145	130	138	156

Recognition

Ages 10 through 11

Raw scores to standard scores Ages 10 through 11					
Recognition Raw	Spoken Words	Written Words	Objects	Designs	Pseudo-words
-30	42	42	36	63	68
-29	42	43	37	63	69
-28	43	43	38	64	69
-27	43	44	38	64	69
-26	44	44	39	65	70
-25	44	45	39	65	70
-24	45	45	40	66	70
-23	45	46	40	66	71
-22	46	46	41	66	71
-21	46	47	42	67	71
-20	47	47	42	67	72
-19	47	48	43	68	72
-18	48	48	43	68	72
-17	48	49	44	69	73
-16	49	49	44	69	73
-15	49	50	45	69	73
-14	50	50	45	70	74
-13	51	51	46	70	74
-12	51	51	47	71	74
-11	52	52	47	71	75
-10	52	52	48	72	75
-9	53	53	48	72	75
-8	53	53	49	73	76
-7	54	54	49	73	76
-6	54	54	50	73	76
-5	55	55	51	74	77
-4	55	55	51	74	77
-3	56	56	52	75	77
-2	56	56	52	75	78
-1	57	57	53	76	78

Raw scores to standard scores Ages 10 through 11					
Recognition Raw	Spoken Words	Written Words	Objects	Designs	Pseudo-words
0	57	57	53	76	78
1	58	58	54	76	79
2	58	59	55	77	79
3	59	59	55	77	79
4	60	60	56	78	80
5	60	60	56	78	80
6	61	61	57	79	80
7	61	61	57	79	81
8	62	62	58	79	81
9	62	62	59	80	81
10	63	63	59	80	82
11	63	63	60	81	82
12	64	64	60	81	82
13	64	64	61	82	83
14	65	65	61	82	83
15	65	65	62	82	83
16	66	66	62	83	84
17	66	66	63	83	84
18	67	67	64	84	84
19	67	67	64	84	85
20	68	68	65	85	85
21	69	68	65	85	85
22	69	69	66	85	86
23	70	69	66	86	86
24	70	70	67	86	86
25	71	70	68	87	87
26	71	71	68	87	87
27	72	71	69	88	87
28	72	72	69	88	88
29	73	72	70	89	88

Recognition

Ages 10 through 11

Raw scores to standard scores Ages 10 through 11					
Recognition Raw	Spoken Words	Written Words	Objects	Designs	Pseudo-words
30	73	73	70	89	88
31	74	73	71	89	89
32	74	74	72	90	89
33	75	75	72	90	89
34	75	75	73	91	90
35	76	76	73	91	90
36	76	76	74	92	90
37	77	77	74	92	91
38	77	77	75	92	91
39	78	78	76	93	91
40	79	78	76	93	92
41	79	79	77	94	92
42	80	79	77	94	92
43	80	80	78	95	93
44	81	80	78	95	93
45	81	81	79	95	93
46	82	81	79	96	94
47	82	82	80	96	94
48	83	82	81	97	94
49	83	83	81	97	95
50	84	83	82	98	95
51	84	84	82	98	95
52	85	84	83	98	96
53	85	85	83	99	96
54	86	85	84	99	96
55	86	86	85	100	97
56	87	86	85	100	97
57	88	87	86	101	97
58	88	87	86	101	98
59	89	88	87	101	98

Raw scores to standard scores Ages 10 through 11					
Recognition Raw	Spoken Words	Written Words	Objects	Designs	Pseudo-words
60	89	88	87	102	98
61	90	89	88	102	98
62	90	89	89	103	99
63	91	90	89	103	99
64	91	91	90	104	99
65	92	91	90	104	100
66	92	92	91	105	100
67	93	92	91	105	100
68	93	93	92	105	101
69	94	93	93	106	101
70	94	94	93	106	101
71	95	94	94	107	102
72	95	95	94	107	102
73	96	95	95	108	102
74	97	96	95	108	103
75	97	96	96	108	103
76	98	97	96	109	103
77	98	97	97	109	104
78	99	98	98	110	104
79	99	98	98	110	104
80	100	99	99	111	105
81	100	99	99	111	105
82	101	100	100	111	105
83	101	100	100	112	106
84	102	101	101	112	106
85	102	101	102	113	106
86	103	102	102	113	107
87	103	102	103	114	107
88	104	103	103	114	107
89	104	103	104	114	108

Raw scores to standard scores Ages 10 through 11					
Recognition Raw	Spoken Words	Written Words	Objects	Designs	Pseudo- words
90	105	104	104	115	108
91	106	104	105	115	108
92	106	105	106	116	109
93	107	105	106	116	109
94	107	106	107	117	109
95	108	107	107	117	110
96	108	107	108	117	110
97	109	108	108	118	110
98	109	108	109	118	111
99	110	109	110	119	111
100	110	109	110	119	111

Raw scores to standard scores Ages 12 through 13					
Learning Raw	Spoken Words	Written Words	Objects	Designs	Pseudo-words
1	55	59	60	61	73
2	56	60	61	61	73
3	56	60	62	62	74
4	57	61	62	63	75
5	58	62	63	64	76
6	59	62	64	65	77
7	60	63	64	66	77
8	60	63	65	67	78
9	61	64	66	68	79
10	62	65	66	69	80
11	63	65	67	70	81
12	63	66	67	71	81
13	64	66	68	72	82
14	65	67	69	73	83
15	66	68	69	74	84
16	67	68	70	75	85
17	67	69	71	76	86
18	68	69	71	77	86
19	69	70	72	78	87
20	70	71	73	78	88
21	71	71	73	79	89
22	71	72	74	80	90
23	72	72	74	81	90
24	73	73	75	82	91
25	74	74	76	83	92
26	75	74	76	84	93
27	75	75	77	85	94
28	76	75	78	86	95
29	77	76	78	87	95
30	78	77	79	88	96

Raw scores to standard scores Ages 12 through 13					
Learning Raw	Spoken Words	Written Words	Objects	Designs	Pseudo-words
31	79	77	80	89	97
32	79	78	80	90	98
33	80	79	81	91	99
34	81	79	81	92	99
35	82	80	82	93	100
36	83	80	83	94	101
37	83	81	83	95	102
38	84	82	84	95	103
39	85	82	85	96	103
40	86	83	85	97	104
41	87	83	86	98	105
42	87	84	87	99	106
43	88	85	87	100	107
44	89	85	88	101	108
45	90	86	88	102	108
46	90	86	89	103	109
47	91	87	90	104	110
48	92	88	90	105	111
49	93	88	91	106	112
50	94	89	92	107	112
51	94	89	92	108	113
52	95	90	93	109	114
53	96	91	94	110	115
54	97	91	94	111	116
55	98	92	95	112	116
56	98	92	95	112	117
57	99	93	96	113	118
58	100	94	97	114	119
59	101	94	97	115	120
60	102	95	98	116	121

Raw scores to standard scores Ages 12 through 13					
Learning Raw	Spoken Words	Written Words	Objects	Designs	Pseudo-words
61	102	96	99	117	121
62	103	96	99	118	122
63	104	97	100	119	123
64	105	97	101	120	124
65	106	98	101	121	125
66	106	99	102	122	125
67	107	99	102	123	126
68	108	100	103	124	127
69	109	100	104	125	128
70	110	101	104	126	129
71	110	102	105	127	129
72	111	102	106	128	130
73	112	103	106	129	131
74	113	103	107	130	132
75	114	104	108	130	133
76	114	105	108	131	134
77	115	105	109	132	134
78	116	106	109	133	135
79	117	106	110	134	136
80	117	107	111	135	137
81	118	108	111	136	138
82	119	108	112	137	138
83	120	109	113	138	139
84	121	109	113	139	140
85	121	110	114	140	141
86	122	111	115	141	142
87	123	111	115	142	142
88	124	112	116	143	143
89	125	113	116	144	144
90	125	113	117	145	145

Raw scores to standard scores Ages 12 through 13					
Learning Raw	Spoken Words	Written Words	Objects	Designs	Pseudo-words
91	126	114	118	146	146
92	127	114	118	147	147
93	128	115	119	147	147
94	129	116	120	148	148
95	129	116	120	149	149
96	130	117	121	150	150
97	131	117	122	151	151
98	132	118	122	152	151
99	133	119	123	153	152
100	133	119	124	154	153

Raw scores to standard scores Ages 12 through 13					
Learning gain Raw	Spoken Words	Written Words	Objects	Designs	Pseudo- words
-30	42	66	47	57	56
-29	43	66	48	58	57
-28	44	67	49	59	58
-27	45	68	50	59	58
-26	46	68	50	60	59
-25	47	69	51	61	60
-24	48	69	52	62	61
-23	48	70	53	63	62
-22	49	70	53	64	62
-21	50	71	54	64	63
-20	51	71	55	65	64
-19	52	72	56	66	65
-18	53	72	57	67	65
-17	54	73	57	68	66
-16	54	74	58	68	67
-15	55	74	59	69	68
-14	56	75	60	70	69
-13	57	75	60	71	69
-12	58	76	61	72	70
-11	59	76	62	73	71
-10	60	77	63	73	72
-9	60	77	64	74	72
-8	61	78	64	75	73
-7	62	78	65	76	74
-6	63	79	66	77	75
-5	64	80	67	77	76
-4	65	80	68	78	76
-3	66	81	68	79	77
-2	66	81	69	80	78
-1	67	82	70	81	79

Raw scores to standard scores Ages 12 through 13					
Learning gain Raw	Spoken Words	Written Words	Objects	Designs	Pseudo- words
0	68	82	71	81	79
1	69	83	71	82	80
2	70	83	72	83	81
3	71	84	73	84	82
4	72	84	74	85	83
5	72	85	75	86	83
6	73	86	75	86	84
7	74	86	76	87	85
8	75	87	77	88	86
9	76	87	78	89	86
10	77	88	78	90	87
11	78	88	79	90	88
12	79	89	80	91	89
13	79	89	81	92	90
14	80	90	82	93	90
15	81	90	82	94	91
16	82	91	83	95	92
17	83	92	84	95	93
18	84	92	85	96	93
19	85	93	85	97	94
20	85	93	86	98	95
21	86	94	87	99	96
22	87	94	88	99	97
23	88	95	89	100	97
24	89	95	89	101	98
25	90	96	90	102	99
26	91	96	91	103	100
27	91	97	92	104	100
28	92	98	93	104	101
29	93	98	93	105	102

Learning Gain

Ages 12 through 13

Raw scores to standard scores Ages 12 through 13					
Learning gain Raw	Spoken Words	Written Words	Objects	Designs	Pseudo- words
30	94	99	94	106	103
31	95	99	95	107	104
32	96	100	96	108	104
33	97	100	96	108	105
34	97	101	97	109	106
35	98	101	98	110	107
36	99	102	99	111	107
37	100	103	100	112	108
38	101	103	100	113	109
39	102	104	101	113	110
40	103	104	102	114	111
41	103	105	103	115	111
42	104	105	103	116	112
43	105	106	104	117	113
44	106	106	105	117	114
45	107	107	106	118	114
46	108	107	107	119	115
47	109	108	107	120	116
48	109	109	108	121	117
49	110	109	109	122	118
50	111	110	110	122	118
51	112	110	110	123	119
52	113	111	111	124	120
53	114	111	112	125	121
54	115	112	113	126	121
55	115	112	114	126	122
56	116	113	114	127	123
57	117	113	115	128	124
58	118	114	116	129	125
59	119	115	117	130	125

Raw scores to standard scores Ages 12 through 13					
Learning gain Raw	Spoken Words	Written Words	Objects	Designs	Pseudo- words
60	120	115	118	130	126
61	121	116	118	131	127
62	121	116	119	132	128
63	122	117	120	133	128
64	123	117	121	134	129
65	124	118	121	135	130
66	125	118	122	135	131
67	126	119	123	136	132
68	127	119	124	137	132
69	128	120	125	138	133
70	128	121	125	139	134
71	129	121	126	139	135
72	130	122	127	140	135
73	131	122	128	141	136
74	132	123	128	142	137
75	133	123	129	143	138
76	134	124	130	144	139
77	134	124	131	144	139
78	135	125	132	145	140
79	136	125	132	146	141

Delayed recall

Ages 12 through 13

Raw scores to standard scores Ages 12 through 13					
Delayed recall	Spoken Words	Written Words	Objects	Designs	Pseudo-words
Raw					
0	55	53	56	68	75
1	56	53	57	69	76
2	57	54	57	69	77
3	57	54	58	70	77
4	58	55	58	71	78
5	59	56	59	71	78
6	59	56	60	72	79
7	60	57	60	73	80
8	60	57	61	73	80
9	61	58	61	74	81
10	62	59	62	75	81
11	62	59	63	75	82
12	63	60	63	76	82
13	64	60	64	77	83
14	64	61	64	77	84
15	65	62	65	78	84
16	66	62	66	79	85
17	66	63	66	79	85
18	67	63	67	80	86
19	67	64	67	81	87
20	68	64	68	81	87
21	69	65	69	82	88
22	69	66	69	82	88
23	70	66	70	83	89
24	71	67	70	84	90
25	71	67	71	84	90
26	72	68	71	85	91
27	72	69	72	86	91
28	73	69	73	86	92
29	74	70	73	87	93

Raw scores to standard scores Ages 12 through 13					
Delayed recall	Spoken Words	Written Words	Objects	Designs	Pseudo-words
Raw					
30	74	70	74	88	93
31	75	71	74	88	94
32	76	72	75	89	94
33	76	72	76	90	95
34	77	73	76	90	96
35	77	73	77	91	96
36	78	74	77	92	97
37	79	75	78	92	97
38	79	75	79	93	98
39	80	76	79	94	99
40	81	76	80	94	99
41	81	77	80	95	100
42	82	77	81	95	100
43	83	78	82	96	101
44	83	79	82	97	102
45	84	79	83	97	102
46	84	80	83	98	103
47	85	80	84	99	103
48	86	81	84	99	104
49	86	82	85	100	105
50	87	82	86	101	105
51	88	83	86	101	106
52	88	83	87	102	106
53	89	84	87	103	107
54	89	85	88	103	107
55	90	85	89	104	108
56	91	86	89	105	109
57	91	86	90	105	109
58	92	87	90	106	110
59	93	88	91	106	110

Delayed recall

Ages 12 through 13

Raw scores to standard scores Ages 12 through 13					
Delayed recall Raw	Spoken Words	Written Words	Objects	Designs	Pseudo- words
60	93	88	92	107	111
61	94	89	92	108	112
62	95	89	93	108	112
63	95	90	93	109	113
64	96	91	94	110	113
65	96	91	94	110	114
66	97	92	95	111	115
67	98	92	96	112	115
68	98	93	96	112	116
69	99	93	97	113	116
70	100	94	97	114	117
71	100	95	98	114	118
72	101	95	99	115	118
73	101	96	99	116	119
74	102	96	100	116	119
75	103	97	100	117	120
76	103	98	101	118	121
77	104	98	102	118	121
78	105	99	102	119	122
79	105	99	103	119	122
80	106	100	103	120	123
81	107	101	104	121	124
82	107	101	105	121	124
83	108	102	105	122	125
84	108	102	106	123	125
85	109	103	106	123	126
86	110	104	107	124	127
87	110	104	107	125	127
88	111	105	108	125	128
89	112	105	109	126	128

Raw scores to standard scores Ages 12 through 13					
Delayed recall Raw	Spoken Words	Written Words	Objects	Designs	Pseudo- words
90	112	106	109	127	129
91	113	107	110	127	130
92	113	107	110	128	130
93	114	108	111	129	131
94	115	108	112	129	131
95	115	109	112	130	132
96	116	109	113	131	132
97	117	110	113	131	133
98	117	111	114	132	134
99	118	111	115	132	134
100	119	112	115	133	135

Forgetting

Ages 12 through 13

Raw scores to standard scores Ages 12 through 13					
Forgetting Raw	Spoken Words	Written Words	Objects	Designs	Pseudo-words
-50	42	47	60	58	59
-49	43	48	61	59	60
-48	44	49	62	60	61
-47	46	50	63	61	62
-46	47	51	63	62	63
-45	48	52	64	63	64
-44	49	54	65	63	65
-43	51	55	66	64	66
-42	52	56	67	65	66
-41	53	57	68	66	67
-40	55	58	69	67	68
-39	56	59	70	68	69
-38	57	60	71	69	70
-37	58	61	71	70	71
-36	60	62	72	71	72
-35	61	63	73	72	73
-34	62	65	74	72	74
-33	64	66	75	73	75
-32	65	67	76	74	76
-31	66	68	77	75	77
-30	67	69	78	76	78
-29	69	70	78	77	79
-28	70	71	79	78	80
-27	71	72	80	79	81
-26	73	73	81	80	82
-25	74	74	82	81	83
-24	75	76	83	81	84
-23	76	77	84	82	85
-22	78	78	85	83	86
-21	79	79	86	84	87

Raw scores to standard scores Ages 12 through 13					
Forgetting Raw	Spoken Words	Written Words	Objects	Designs	Pseudo-words
-20	80	80	86	85	88
-19	82	81	87	86	89
-18	83	82	88	87	90
-17	84	83	89	88	91
-16	85	84	90	89	91
-15	87	85	91	90	92
-14	88	87	92	90	93
-13	89	88	93	91	94
-12	91	89	94	92	95
-11	92	90	94	93	96
-10	93	91	95	94	97
-9	94	92	96	95	98
-8	96	93	97	96	99
-7	97	94	98	97	100
-6	98	95	99	98	101
-5	100	96	100	99	102
-4	101	97	101	99	103
-3	102	99	101	100	104
-2	103	100	102	101	105
-1	105	101	103	102	106
0	106	102	104	103	107
1	107	103	105	104	108
2	109	104	106	105	109
3	110	105	107	106	110
4	111	106	108	107	111
5	112	107	109	107	112
6	114	108	109	108	113
7	115	110	110	109	114
8	116	111	111	110	115
9	118	112	112	111	116

Raw scores to standard scores Ages 12 through 13					
Forgetting Raw	Spoken Words	Written Words	Objects	Designs	Pseudo- words
10	119	113	113	112	116
11	120	114	114	113	117
12	121	115	115	114	118
13	123	116	116	115	119
14	124	117	117	116	120
15	125	118	117	116	121
16	127	119	118	117	122
17	128	121	119	118	123
18	129	122	120	119	124
19	130	123	121	120	125
20	132	124	122	121	126
21	133	125	123	122	127
22	134	126	124	123	128
23	136	127	124	124	129
24	137	128	125	125	130
25	138	129	126	125	131

Recognition

Ages 12 through 13

Raw scores to standard scores Ages 12 through 13					
Recognition Raw	Spoken Words	Written Words	Objects	Designs	Pseudo- words
-30	15	1	13	35	48
-29	16	1	14	36	49
-28	16	1	14	36	49
-27	17	1	15	37	50
-26	18	1	16	37	50
-25	19	1	17	38	51
-24	19	1	17	39	51
-23	20	1	18	39	52
-22	21	1	19	40	52
-21	21	1	20	40	53
-20	22	1	20	41	53
-19	23	1	21	42	54
-18	24	1	22	42	54
-17	24	1	22	43	55
-16	25	1	23	43	55
-15	26	2	24	44	55
-14	27	3	25	44	56
-13	27	4	25	45	56
-12	28	5	26	46	57
-11	29	6	27	46	57
-10	30	7	28	47	58
-9	30	8	28	47	58
-8	31	9	29	48	59
-7	32	10	30	49	59
-6	32	11	31	49	60
-5	33	12	31	50	60
-4	34	13	32	50	61
-3	35	13	33	51	61
-2	35	14	34	52	62
-1	36	15	34	52	62

Raw scores to standard scores Ages 12 through 13					
Recognition Raw	Spoken Words	Written Words	Objects	Designs	Pseudo- words
0	37	16	35	53	63
1	38	17	36	53	63
2	38	18	36	54	64
3	39	19	37	55	64
4	40	20	38	55	65
5	41	21	39	56	65
6	41	22	39	56	66
7	42	23	40	57	66
8	43	24	41	58	67
9	44	24	42	58	67
10	44	25	42	59	68
11	45	26	43	59	68
12	46	27	44	60	69
13	46	28	45	61	69
14	47	29	45	61	70
15	48	30	46	62	70
16	49	31	47	62	71
17	49	32	48	63	71
18	50	33	48	64	72
19	51	34	49	64	72
20	52	35	50	65	73
21	52	35	50	65	73
22	53	36	51	66	74
23	54	37	52	67	74
24	55	38	53	67	74
25	55	39	53	68	75
26	56	40	54	68	75
27	57	41	55	69	76
28	57	42	56	70	76
29	58	43	56	70	77

Raw scores to standard scores Ages 12 through 13					
Recognition Raw	Spoken Words	Written Words	Objects	Designs	Pseudo- words
30	59	44	57	71	77
31	60	45	58	71	78
32	60	46	59	72	78
33	61	46	59	73	79
34	62	47	60	73	79
35	63	48	61	74	80
36	63	49	61	74	80
37	64	50	62	75	81
38	65	51	63	76	81
39	66	52	64	76	82
40	66	53	64	77	82
41	67	54	65	77	83
42	68	55	66	78	83
43	68	56	67	79	84
44	69	57	67	79	84
45	70	57	68	80	85
46	71	58	69	80	85
47	71	59	70	81	86
48	72	60	70	82	86
49	73	61	71	82	87
50	74	62	72	83	87
51	74	63	73	83	88
52	75	64	73	84	88
53	76	65	74	85	89
54	77	66	75	85	89
55	77	67	75	86	90
56	78	68	76	86	90
57	79	68	77	87	91
58	79	69	78	88	91
59	80	70	78	88	92

Raw scores to standard scores Ages 12 through 13					
Recognition Raw	Spoken Words	Written Words	Objects	Designs	Pseudo- words
60	81	71	79	89	92
61	82	72	80	89	92
62	82	73	81	90	93
63	83	74	81	91	93
64	84	75	82	91	94
65	85	76	83	92	94
66	85	77	84	92	95
67	86	78	84	93	95
68	87	79	85	94	96
69	88	80	86	94	96
70	88	80	87	95	97
71	89	81	87	95	97
72	90	82	88	96	98
73	90	83	89	97	98
74	91	84	89	97	99
75	92	85	90	98	99
76	93	86	91	98	100
77	93	87	92	99	100
78	94	88	92	100	101
79	95	89	93	100	101
80	96	90	94	101	102
81	96	91	95	101	102
82	97	91	95	102	103
83	98	92	96	103	103
84	99	93	97	103	104
85	99	94	98	104	104
86	100	95	98	104	105
87	101	96	99	105	105
88	101	97	100	106	106
89	102	98	101	106	106

Raw scores to standard scores Ages 12 through 13					
Recognition Raw	Spoken Words	Written Words	Objects	Designs	Pseudo- words
90	103	99	101	107	107
91	104	100	102	107	107
92	104	101	103	108	108
93	105	102	103	109	108
94	106	102	104	109	109
95	107	103	105	110	109
96	107	104	106	110	110
97	108	105	106	111	110
98	109	106	107	112	110
99	110	107	108	112	111
100	110	108	109	113	111

Raw scores to standard scores Ages 14 through 15					
Learning Raw	Spoken Words	Written Words	Objects	Designs	Pseudo-words
1	63	61	67	65	75
2	64	61	67	66	76
3	64	62	68	66	77
4	65	63	68	67	77
5	66	63	69	68	78
6	66	64	70	68	78
7	67	64	70	69	79
8	68	65	71	70	80
9	68	66	71	70	80
10	69	66	72	71	81
11	70	67	72	72	81
12	70	67	73	72	82
13	71	68	74	73	83
14	72	68	74	74	83
15	72	69	75	75	84
16	73	70	75	75	84
17	74	70	76	76	85
18	74	71	76	77	86
19	75	71	77	77	86
20	76	72	78	78	87
21	76	73	78	79	87
22	77	73	79	79	88
23	78	74	79	80	89
24	78	74	80	81	89
25	79	75	80	81	90
26	80	75	81	82	90
27	80	76	82	83	91
28	81	77	82	83	92
29	82	77	83	84	92
30	82	78	83	85	93

Raw scores to standard scores Ages 14 through 15					
Learning Raw	Spoken Words	Written Words	Objects	Designs	Pseudo-words
31	83	78	84	85	93
32	84	79	85	86	94
33	84	80	85	87	95
34	85	80	86	87	95
35	86	81	86	88	96
36	86	81	87	89	96
37	87	82	87	89	97
38	88	82	88	90	98
39	88	83	89	91	98
40	89	84	89	91	99
41	90	84	90	92	99
42	91	85	90	93	100
43	91	85	91	93	101
44	92	86	91	94	101
45	93	86	92	95	102
46	93	87	93	96	102
47	94	88	93	96	103
48	95	88	94	97	104
49	95	89	94	98	104
50	96	89	95	98	105
51	97	90	96	99	105
52	97	91	96	100	106
53	98	91	97	100	107
54	99	92	97	101	107
55	99	92	98	102	108
56	100	93	98	102	108
57	101	93	99	103	109
58	101	94	100	104	109
59	102	95	100	104	110
60	103	95	101	105	111

Raw scores to standard scores Ages 14 through 15					
Learning Raw	Spoken Words	Written Words	Objects	Designs	Pseudo-words
61	103	96	101	106	111
62	104	96	102	106	112
63	105	97	102	107	112
64	105	98	103	108	113
65	106	98	104	108	114
66	107	99	104	109	114
67	107	99	105	110	115
68	108	100	105	110	115
69	109	100	106	111	116
70	109	101	106	112	117
71	110	102	107	112	117
72	111	102	108	113	118
73	111	103	108	114	118
74	112	103	109	114	119
75	113	104	109	115	120
76	113	105	110	116	120
77	114	105	111	117	121
78	115	106	111	117	121
79	115	106	112	118	122
80	116	107	112	119	123
81	117	107	113	119	123
82	117	108	113	120	124
83	118	109	114	121	124
84	119	109	115	121	125
85	119	110	115	122	126
86	120	110	116	123	126
87	121	111	116	123	127
88	121	111	117	124	127
89	122	112	117	125	128
90	123	113	118	125	129

Raw scores to standard scores Ages 14 through 15					
Learning Raw	Spoken Words	Written Words	Objects	Designs	Pseudo-words
91	123	113	119	126	129
92	124	114	119	127	130
93	125	114	120	127	130
94	125	115	120	128	131
95	126	116	121	129	132
96	127	116	122	129	132
97	127	117	122	130	133
98	128	117	123	131	133
99	129	118	123	131	134
100	129	118	124	132	135

Raw scores to standard scores Ages 14 through 15					
Learning gain Raw	Spoken Words	Written Words	Objects	Designs	Pseudo- words
-30	47	62	47	55	51
-29	48	63	47	56	52
-28	49	63	48	57	53
-27	49	64	49	58	53
-26	50	64	50	59	54
-25	51	65	51	59	55
-24	52	65	52	60	56
-23	53	66	52	61	57
-22	53	67	53	62	57
-21	54	67	54	63	58
-20	55	68	55	64	59
-19	56	68	56	64	60
-18	57	69	57	65	60
-17	57	69	57	66	61
-16	58	70	58	67	62
-15	59	71	59	68	63
-14	60	71	60	69	64
-13	60	72	61	69	64
-12	61	72	62	70	65
-11	62	73	62	71	66
-10	63	73	63	72	67
-9	64	74	64	73	68
-8	64	74	65	74	68
-7	65	75	66	74	69
-6	66	76	67	75	70
-5	67	76	67	76	71
-4	68	77	68	77	72
-3	68	77	69	78	72
-2	69	78	70	79	73
-1	70	78	71	79	74

Raw scores to standard scores Ages 14 through 15					
Learning gain Raw	Spoken Words	Written Words	Objects	Designs	Pseudo- words
0	71	79	72	80	75
1	71	80	72	81	75
2	72	80	73	82	76
3	73	81	74	83	77
4	74	81	75	84	78
5	75	82	76	84	79
6	75	82	77	85	79
7	76	83	77	86	80
8	77	83	78	87	81
9	78	84	79	88	82
10	79	85	80	89	83
11	79	85	81	89	83
12	80	86	82	90	84
13	81	86	82	91	85
14	82	87	83	92	86
15	82	87	84	93	87
16	83	88	85	94	87
17	84	89	86	94	88
18	85	89	87	95	89
19	86	90	87	96	90
20	86	90	88	97	90
21	87	91	89	98	91
22	88	91	90	99	92
23	89	92	91	99	93
24	90	92	92	100	94
25	90	93	92	101	94
26	91	94	93	102	95
27	92	94	94	103	96
28	93	95	95	104	97
29	93	95	96	104	98

Learning Gain

Ages 14 through 15

Raw scores to standard scores Ages 14 through 15					
Learning gain Raw	Spoken Words	Written Words	Objects	Designs	Pseudo- words
30	94	96	97	105	98
31	95	96	97	106	99
32	96	97	98	107	100
33	97	98	99	108	101
34	97	98	100	109	101
35	98	99	101	109	102
36	99	99	101	110	103
37	100	100	102	111	104
38	100	100	103	112	105
39	101	101	104	113	105
40	102	101	105	114	106
41	103	102	106	114	107
42	104	103	106	115	108
43	104	103	107	116	109
44	105	104	108	117	109
45	106	104	109	118	110
46	107	105	110	119	111
47	108	105	111	119	112
48	108	106	111	120	113
49	109	107	112	121	113
50	110	107	113	122	114
51	111	108	114	123	115
52	111	108	115	124	116
53	112	109	116	124	116
54	113	109	116	125	117
55	114	110	117	126	118
56	115	110	118	127	119
57	115	111	119	128	120
58	116	112	120	129	120
59	117	112	121	129	121

Raw scores to standard scores Ages 14 through 15					
Learning gain Raw	Spoken Words	Written Words	Objects	Designs	Pseudo- words
60	118	113	121	130	122
61	119	113	122	131	123
62	119	114	123	132	124
63	120	114	124	133	124
64	121	115	125	134	125
65	122	116	126	134	126
66	122	116	126	135	127
67	123	117	127	136	128
68	124	117	128	137	128
69	125	118	129	138	129
70	126	118	130	139	130

Delayed recall

Ages 14 through 15

Raw scores to standard scores Ages 14 through 15					
Delayed recall	Spoken Words	Written Words	Objects	Designs	Pseudo-words
Raw					
0	64	53	64	62	76
1	65	54	64	63	76
2	65	54	65	64	77
3	66	55	65	64	77
4	66	55	66	65	78
5	67	56	66	65	78
6	67	57	67	66	79
7	68	57	67	67	79
8	68	58	68	67	79
9	69	58	68	68	80
10	69	59	69	69	80
11	70	59	69	69	81
12	70	60	70	70	81
13	71	61	70	71	82
14	71	61	71	71	82
15	72	62	71	72	83
16	72	62	72	72	83
17	73	63	72	73	84
18	74	64	73	74	84
19	74	64	73	74	85
20	75	65	74	75	85
21	75	65	75	76	86
22	76	66	75	76	86
23	76	66	76	77	87
24	77	67	76	78	87
25	77	68	77	78	87
26	78	68	77	79	88
27	78	69	78	79	88
28	79	69	78	80	89
29	79	70	79	81	89

Raw scores to standard scores Ages 14 through 15					
Delayed recall	Spoken Words	Written Words	Objects	Designs	Pseudo-words
Raw					
30	80	71	79	81	90
31	80	71	80	82	90
32	81	72	80	83	91
33	81	72	81	83	91
34	82	73	81	84	92
35	82	73	82	84	92
36	83	74	82	85	93
37	83	75	83	86	93
38	84	75	83	86	94
39	84	76	84	87	94
40	85	76	84	88	95
41	85	77	85	88	95
42	86	78	85	89	95
43	86	78	86	90	96
44	87	79	86	90	96
45	87	79	87	91	97
46	88	80	87	91	97
47	88	80	88	92	98
48	89	81	88	93	98
49	89	82	89	93	99
50	90	82	89	94	99
51	90	83	90	95	100
52	91	83	90	95	100
53	92	84	91	96	101
54	92	85	91	97	101
55	93	85	92	97	102
56	93	86	93	98	102
57	94	86	93	98	103
58	94	87	94	99	103
59	95	87	94	100	103

Delayed recall

Ages 14 through 15

Raw scores to standard scores Ages 14 through 15					
Delayed recall	Spoken Words	Written Words	Objects	Designs	Pseudo-words
Raw					
60	95	88	95	100	104
61	96	89	95	101	104
62	96	89	96	102	105
63	97	90	96	102	105
64	97	90	97	103	106
65	98	91	97	104	106
66	98	92	98	104	107
67	99	92	98	105	107
68	99	93	99	105	108
69	100	93	99	106	108
70	100	94	100	107	109
71	101	94	100	107	109
72	101	95	101	108	110
73	102	96	101	109	110
74	102	96	102	109	111
75	103	97	102	110	111
76	103	97	103	111	112
77	104	98	103	111	112
78	104	99	104	112	112
79	105	99	104	112	113
80	105	100	105	113	113
81	106	100	105	114	114
82	106	101	106	114	114
83	107	101	106	115	115
84	107	102	107	116	115
85	108	103	107	116	116
86	109	103	108	117	116
87	109	104	108	117	117
88	110	104	109	118	117
89	110	105	110	119	118

Raw scores to standard scores Ages 14 through 15					
Delayed recall	Spoken Words	Written Words	Objects	Designs	Pseudo-words
Raw					
90	111	106	110	119	118
91	111	106	111	120	119
92	112	107	111	121	119
93	112	107	112	121	120
94	113	108	112	122	120
95	113	108	113	123	120
96	114	109	113	123	121
97	114	110	114	124	121
98	115	110	114	124	122
99	115	111	115	125	122
100	116	111	115	126	123

Recognition

Ages 14 through 15

Raw scores to standard scores Ages 14 through 15					
Recognition Raw	Spoken Words	Written Words	Objects	Designs	Pseudo- words
-30	18	22	29	23	66
-29	19	23	29	24	67
-28	19	23	30	24	67
-27	20	24	30	25	67
-26	21	25	31	26	68
-25	21	25	32	26	68
-24	22	26	32	27	68
-23	23	27	33	28	69
-22	24	27	33	28	69
-21	24	28	34	29	69
-20	25	28	35	30	70
-19	26	29	35	30	70
-18	26	30	36	31	70
-17	27	30	36	32	71
-16	28	31	37	32	71
-15	28	32	38	33	71
-14	29	32	38	34	72
-13	30	33	39	34	72
-12	30	34	39	35	72
-11	31	34	40	36	73
-10	32	35	41	36	73
-9	32	36	41	37	73
-8	33	36	42	38	74
-7	34	37	42	38	74
-6	34	37	43	39	75
-5	35	38	44	40	75
-4	36	39	44	40	75
-3	36	39	45	41	76
-2	37	40	45	42	76
-1	38	41	46	42	76

Raw scores to standard scores Ages 14 through 15					
Recognition Raw	Spoken Words	Written Words	Objects	Designs	Pseudo- words
0	39	41	47	43	77
1	39	42	47	44	77
2	40	43	48	44	77
3	41	43	48	45	78
4	41	44	49	46	78
5	42	44	50	46	78
6	43	45	50	47	79
7	43	46	51	48	79
8	44	46	51	48	79
9	45	47	52	49	80
10	45	48	53	50	80
11	46	48	53	50	80
12	47	49	54	51	81
13	47	50	54	52	81
14	48	50	55	52	81
15	49	51	56	53	82
16	49	52	56	54	82
17	50	52	57	54	82
18	51	53	57	55	83
19	52	53	58	56	83
20	52	54	59	56	83
21	53	55	59	57	84
22	54	55	60	58	84
23	54	56	60	58	84
24	55	57	61	59	85
25	56	57	62	60	85
26	56	58	62	60	85
27	57	59	63	61	86
28	58	59	63	62	86
29	58	60	64	62	86

Recognition

Ages 14 through 15

Raw scores to standard scores Ages 14 through 15					
Recognition Raw	Spoken Words	Written Words	Objects	Designs	Pseudo- words
30	59	61	65	63	87
31	60	61	65	64	87
32	60	62	66	64	87
33	61	62	66	65	88
34	62	63	67	66	88
35	62	64	68	66	88
36	63	64	68	67	89
37	64	65	69	68	89
38	64	66	69	68	89
39	65	66	70	69	90
40	66	67	71	70	90
41	67	68	71	70	90
42	67	68	72	71	91
43	68	69	72	72	91
44	69	69	73	72	91
45	69	70	74	73	92
46	70	71	74	74	92
47	71	71	75	74	92
48	71	72	75	75	93
49	72	73	76	76	93
50	73	73	77	76	93
51	73	74	77	77	94
52	74	75	78	78	94
53	75	75	78	78	94
54	75	76	79	79	95
55	76	77	80	80	95
56	77	77	80	80	95
57	77	78	81	81	96
58	78	78	81	82	96
59	79	79	82	82	96

Raw scores to standard scores Ages 14 through 15					
Recognition Raw	Spoken Words	Written Words	Objects	Designs	Pseudo- words
60	80	80	83	83	97
61	80	80	83	84	97
62	81	81	84	84	97
63	82	82	84	85	98
64	82	82	85	86	98
65	83	83	86	86	98
66	84	84	86	87	99
67	84	84	87	88	99
68	85	85	87	88	100
69	86	86	88	89	100
70	86	86	89	90	100
71	87	87	89	90	101
72	88	87	90	91	101
73	88	88	90	92	101
74	89	89	91	92	102
75	90	89	92	93	102
76	90	90	92	94	102
77	91	91	93	94	103
78	92	91	93	95	103
79	92	92	94	96	103
80	93	93	95	96	104
81	94	93	95	97	104
82	95	94	96	98	104
83	95	94	96	98	105
84	96	95	97	99	105
85	97	96	98	100	105
86	97	96	98	100	106
87	98	97	99	101	106
88	99	98	99	102	106
89	99	98	100	102	107

Raw scores to standard scores Ages 14 through 15					
Recognition Raw	Spoken Words	Written Words	Objects	Designs	Pseudo- words
90	100	99	101	103	107
91	101	100	101	104	107
92	101	100	102	104	108
93	102	101	102	105	108
94	103	102	103	106	108
95	103	102	104	106	109
96	104	103	104	107	109
97	105	103	105	108	109
98	105	104	105	108	110
99	106	105	106	109	110
100	107	105	107	110	110

Raw scores to standard scores Ages 16 through 18					
Learning Raw	Spoken Words	Written Words	Objects	Designs	Pseudo-words
1	50	45	7	49	63
2	51	46	8	50	64
3	52	46	9	51	65
4	52	47	10	52	66
5	53	48	11	53	66
6	54	49	13	54	67
7	55	49	14	54	68
8	55	50	15	55	68
9	56	51	16	56	69
10	57	52	17	57	70
11	57	52	18	58	71
12	58	53	20	59	71
13	59	54	21	60	72
14	60	55	22	61	73
15	60	55	23	61	74
16	61	56	24	62	74
17	62	57	26	63	75
18	63	57	27	64	76
19	63	58	28	65	76
20	64	59	29	66	77
21	65	60	30	67	78
22	65	60	32	68	79
23	66	61	33	69	79
24	67	62	34	69	80
25	68	63	35	70	81
26	68	63	36	71	81
27	69	64	37	72	82
28	70	65	39	73	83
29	70	66	40	74	84
30	71	66	41	75	84

Raw scores to standard scores Ages 16 through 18					
Learning Raw	Spoken Words	Written Words	Objects	Designs	Pseudo-words
31	72	67	42	76	85
32	73	68	43	77	86
33	73	69	45	77	86
34	74	69	46	78	87
35	75	70	47	79	88
36	75	71	48	80	89
37	76	71	49	81	89
38	77	72	50	82	90
39	78	73	52	83	91
40	78	74	53	84	91
41	79	74	54	84	92
42	80	75	55	85	93
43	81	76	56	86	94
44	81	77	58	87	94
45	82	77	59	88	95
46	83	78	60	89	96
47	83	79	61	90	97
48	84	80	62	91	97
49	85	80	63	92	98
50	86	81	65	92	99
51	86	82	66	93	99
52	87	83	67	94	100
53	88	83	68	95	101
54	88	84	69	96	102
55	89	85	71	97	102
56	90	85	72	98	103
57	91	86	73	99	104
58	91	87	74	100	104
59	92	88	75	100	105
60	93	88	77	101	106

Raw scores to standard scores Ages 16 through 18					
Learning Raw	Spoken Words	Written Words	Objects	Designs	Pseudo-words
61	93	89	78	102	107
62	94	90	79	103	107
63	95	91	80	104	108
64	96	91	81	105	109
65	96	92	82	106	109
66	97	93	84	107	110
67	98	94	85	107	111
68	99	94	86	108	112
69	99	95	87	109	112
70	100	96	88	110	113
71	101	97	90	111	114
72	101	97	91	112	115
73	102	98	92	113	115
74	103	99	93	114	116
75	104	100	94	115	117
76	104	100	95	115	117
77	105	101	97	116	118
78	106	102	98	117	119
79	106	102	99	118	120
80	107	103	100	119	120
81	108	104	101	120	121
82	109	105	103	121	122
83	109	105	104	122	122
84	110	106	105	123	123
85	111	107	106	123	124
86	111	108	107	124	125
87	112	108	108	125	125
88	113	109	110	126	126
89	114	110	111	127	127
90	114	111	112	128	127

Raw scores to standard scores Ages 16 through 18					
Learning Raw	Spoken Words	Written Words	Objects	Designs	Pseudo-words
91	115	111	113	129	128
92	116	112	114	130	129
93	116	113	116	130	130
94	117	114	117	131	130
95	118	114	118	132	131
96	119	115	119	133	132
97	119	116	120	134	133
98	120	116	121	135	133
99	121	117	123	136	134
100	122	118	124	137	135

Raw scores to standard scores Ages 16 through 18					
Learning gain Raw	Spoken Words	Written Words	Objects	Designs	Pseudo- words
-30	39	53	42	46	50
-29	40	54	43	47	51
-28	41	55	44	48	52
-27	42	56	45	49	52
-26	43	56	46	49	53
-25	44	57	47	50	54
-24	45	58	48	51	55
-23	46	59	48	52	55
-22	47	59	49	53	56
-21	47	60	50	54	57
-20	48	61	51	54	57
-19	49	61	52	55	58
-18	50	62	53	56	59
-17	51	63	54	57	60
-16	52	64	55	58	60
-15	53	64	56	59	61
-14	54	65	57	60	62
-13	55	66	57	60	62
-12	55	67	58	61	63
-11	56	67	59	62	64
-10	57	68	60	63	64
-9	58	69	61	64	65
-8	59	69	62	65	66
-7	60	70	63	66	67
-6	61	71	64	66	67
-5	62	72	65	67	68
-4	63	72	66	68	69
-3	64	73	67	69	69
-2	64	74	67	70	70
-1	65	75	68	71	71

Raw scores to standard scores Ages 16 through 18					
Learning gain Raw	Spoken Words	Written Words	Objects	Designs	Pseudo- words
0	66	75	69	72	72
1	67	76	70	72	72
2	68	77	71	73	73
3	69	77	72	74	74
4	70	78	73	75	74
5	71	79	74	76	75
6	72	80	75	77	76
7	73	80	76	78	76
8	73	81	77	78	77
9	74	82	77	79	78
10	75	83	78	80	79
11	76	83	79	81	79
12	77	84	80	82	80
13	78	85	81	83	81
14	79	85	82	84	81
15	80	86	83	84	82
16	81	87	84	85	83
17	81	88	85	86	84
18	82	88	86	87	84
19	83	89	87	88	85
20	84	90	87	89	86
21	85	91	88	90	86
22	86	91	89	90	87
23	87	92	90	91	88
24	88	93	91	92	89
25	89	93	92	93	89
26	90	94	93	94	90
27	90	95	94	95	91
28	91	96	95	96	91
29	92	96	96	96	92

Raw scores to standard scores Ages 16 through 18					
Learning gain Raw	Spoken Words	Written Words	Objects	Designs	Pseudo- words
30	93	97	97	97	93
31	94	98	97	98	93
32	95	99	98	99	94
33	96	99	99	100	95
34	97	100	100	101	96
35	98	101	101	101	96
36	98	101	102	102	97
37	99	102	103	103	98
38	100	103	104	104	98
39	101	104	105	105	99
40	102	104	106	106	100
41	103	105	107	107	101
42	104	106	107	107	101
43	105	107	108	108	102
44	106	107	109	109	103
45	107	108	110	110	103
46	107	109	111	111	104
47	108	109	112	112	105
48	109	110	113	113	105
49	110	111	114	113	106
50	111	112	115	114	107
51	112	112	116	115	108
52	113	113	117	116	108
53	114	114	117	117	109
54	115	115	118	118	110
55	116	115	119	119	110
56	116	116	120	119	111
57	117	117	121	120	112
58	118	118	122	121	113
59	119	118	123	122	113

Raw scores to standard scores Ages 16 through 18					
Learning gain Raw	Spoken Words	Written Words	Objects	Designs	Pseudo- words
60	120	119	124	123	114
61	121	120	125	124	115
62	122	120	126	125	115
63	123	121	127	125	116
64	124	122	127	126	117
65	124	123	128	127	117
66	125	123	129	128	118
67	126	124	130	129	119
68	127	125	131	130	120
69	128	126	132	131	120
70	129	126	133	131	121

Delayed recall

Ages 16 through 18

Raw scores to standard scores Ages 16 through 18					
Delayed recall	Spoken Words	Written Words	Objects	Designs	Pseudo-words
Raw					
0	41	34	<1	49	64
1	41	35	<1	49	65
2	42	36	<1	50	65
3	43	37	<1	51	66
4	43	37	<1	52	66
5	44	38	<1	52	67
6	45	39	<1	53	67
7	45	40	<1	54	68
8	46	40	<1	54	69
9	47	41	<1	55	69
10	48	42	<1	56	70
11	48	43	<1	56	70
12	49	43	<1	57	71
13	50	44	<1	58	71
14	50	45	<1	59	72
15	51	46	<1	59	73
16	52	46	<1	60	73
17	53	47	<1	61	74
18	53	48	1	61	74
19	54	49	3	62	75
20	55	49	4	63	75
21	55	50	5	64	76
22	56	51	7	64	77
23	57	52	8	65	77
24	57	52	9	66	78
25	58	53	11	66	78
26	59	54	12	67	79
27	60	55	13	68	79
28	60	55	15	68	80
29	61	56	16	69	81

Raw scores to standard scores Ages 16 through 18					
Delayed recall	Spoken Words	Written Words	Objects	Designs	Pseudo-words
Raw					
30	62	57	17	70	81
31	62	58	19	71	82
32	63	58	20	71	82
33	64	59	21	72	83
34	64	60	23	73	83
35	65	61	24	73	84
36	66	61	25	74	85
37	67	62	27	75	85
38	67	63	28	76	86
39	68	64	29	76	86
40	69	64	31	77	87
41	69	65	32	78	87
42	70	66	33	78	88
43	71	67	35	79	89
44	72	68	36	80	89
45	72	68	37	80	90
46	73	69	39	81	90
47	74	70	40	82	91
48	74	71	41	83	91
49	75	71	43	83	92
50	76	72	44	84	93
51	76	73	45	85	93
52	77	74	47	85	94
53	78	74	48	86	94
54	79	75	49	87	95
55	79	76	51	87	95
56	80	77	52	88	96
57	81	77	53	89	97
58	81	78	55	90	97
59	82	79	56	90	98

Raw scores to standard scores Ages 16 through 18					
Delayed recall	Spoken Words	Written Words	Objects	Designs	Pseudo-words
Raw					
60	83	80	57	91	98
61	84	80	59	92	99
62	84	81	60	92	99
63	85	82	61	93	100
64	86	83	63	94	101
65	86	83	64	95	101
66	87	84	65	95	102
67	88	85	67	96	102
68	88	86	68	97	103
69	89	86	69	97	103
70	90	87	71	98	104
71	91	88	72	99	105
72	91	89	73	99	105
73	92	89	75	100	106
74	93	90	76	101	106
75	93	91	77	102	107
76	94	92	79	102	107
77	95	92	80	103	108
78	95	93	81	104	109
79	96	94	83	104	109
80	97	95	84	105	110
81	98	95	85	106	110
82	98	96	87	106	111
83	99	97	88	107	111
84	100	98	89	108	112
85	100	98	91	109	113
86	101	99	92	109	113
87	102	100	93	110	114
88	103	101	95	111	114
89	103	101	96	111	115

Raw scores to standard scores Ages 16 through 18					
Delayed recall	Spoken Words	Written Words	Objects	Designs	Pseudo-words
Raw					
90	104	102	97	112	115
91	105	103	99	113	116
92	105	104	100	114	117
93	106	104	101	114	117
94	107	105	103	115	118
95	107	106	104	116	118
96	108	107	105	116	119
97	109	107	107	117	119
98	110	108	108	118	120
99	110	109	109	118	121
100	111	110	111	119	121

Recognition

Ages 16 through 18

Raw scores to standard scores Ages 16 through 18					
Recognition Raw	Spoken Words	Written Words	Objects	Designs	Pseudo-words
-30	12	20	1	37	1
-29	13	21	1	37	1
-28	13	22	1	38	1
-27	14	22	1	38	1
-26	15	23	1	39	1
-25	15	24	1	39	1
-24	16	24	1	40	1
-23	17	25	1	40	2
-22	18	26	1	41	3
-21	18	26	1	42	4
-20	19	27	1	42	5
-19	20	27	1	43	5
-18	21	28	1	43	6
-17	21	29	1	44	7
-16	22	29	1	44	8
-15	23	30	1	45	9
-14	23	31	1	45	10
-13	24	31	1	46	11
-12	25	32	1	46	12
-11	26	33	1	47	12
-10	26	33	1	48	13
-9	27	34	1	48	14
-8	28	35	1	49	15
-7	28	35	1	49	16
-6	29	36	1	50	17
-5	30	37	1	50	18
-4	31	37	1	51	19
-3	31	38	1	51	20
-2	32	38	1	52	20
-1	33	39	1	53	21

Raw scores to standard scores Ages 16 through 18					
Recognition Raw	Spoken Words	Written Words	Objects	Designs	Pseudo-words
0	34	40	1	53	22
1	34	40	1	54	23
2	35	41	1	54	24
3	36	42	1	55	25
4	36	42	1	55	26
5	37	43	1	56	27
6	38	44	1	56	27
7	39	44	1	57	28
8	39	45	1	57	29
9	40	46	1	58	30
10	41	46	1	59	31
11	42	47	1	59	32
12	42	47	1	60	33
13	43	48	1	60	34
14	44	49	1	61	35
15	44	49	1	61	35
16	45	50	2	62	36
17	46	51	3	62	37
18	47	51	4	63	38
19	47	52	6	64	39
20	48	53	7	64	40
21	49	53	8	65	41
22	49	54	9	65	42
23	50	55	10	66	42
24	51	55	12	66	43
25	52	56	13	67	44
26	52	57	14	67	45
27	53	57	15	68	46
28	54	58	16	68	47
29	55	58	18	69	48

Recognition

Ages 16 through 18

Raw scores to standard scores Ages 16 through 18					
Recognition Raw	Spoken Words	Written Words	Objects	Designs	Pseudo- words
30	55	59	19	70	49
31	56	60	20	70	49
32	57	60	21	71	50
33	57	61	22	71	51
34	58	62	24	72	52
35	59	62	25	72	53
36	60	63	26	73	54
37	60	64	27	73	55
38	61	64	28	74	56
39	62	65	30	75	57
40	62	66	31	75	57
41	63	66	32	76	58
42	64	67	33	76	59
43	65	68	34	77	60
44	65	68	36	77	61
45	66	69	37	78	62
46	67	69	38	78	63
47	68	70	39	79	64
48	68	71	40	80	64
49	69	71	42	80	65
50	70	72	43	81	66
51	70	73	44	81	67
52	71	73	45	82	68
53	72	74	46	82	69
54	73	75	48	83	70
55	73	75	49	83	71
56	74	76	50	84	71
57	75	77	51	84	72
58	75	77	52	85	73
59	76	78	54	86	74

Raw scores to standard scores Ages 16 through 18					
Recognition Raw	Spoken Words	Written Words	Objects	Designs	Pseudo- words
60	77	78	55	86	75
61	78	79	56	87	76
62	78	80	57	87	77
63	79	80	58	88	78
64	80	81	60	88	79
65	81	82	61	89	79
66	81	82	62	89	80
67	82	83	63	90	81
68	83	84	64	91	82
69	83	84	66	91	83
70	84	85	67	92	84
71	85	86	68	92	85
72	86	86	69	93	86
73	86	87	70	93	86
74	87	88	72	94	87
75	88	88	73	94	88
76	89	89	74	95	89
77	89	89	75	95	90
78	90	90	76	96	91
79	91	91	78	97	92
80	91	91	79	97	93
81	92	92	80	98	94
82	93	93	81	98	94
83	94	93	82	99	95
84	94	94	84	99	96
85	95	95	85	100	97
86	96	95	86	100	98
87	96	96	87	101	99
88	97	97	88	102	100
89	98	97	90	102	101

Raw scores to standard scores Ages 16 through 18					
Recognition Raw	Spoken Words	Written Words	Objects	Designs	Pseudo- words
90	99	98	91	103	101
91	99	98	92	103	102
92	100	99	93	104	103
93	101	100	94	104	104
94	102	100	96	105	105
95	102	101	97	105	106
96	103	102	98	106	107
97	104	102	99	106	108
98	104	103	100	107	108
99	105	104	102	108	109
100	106	104	103	108	110